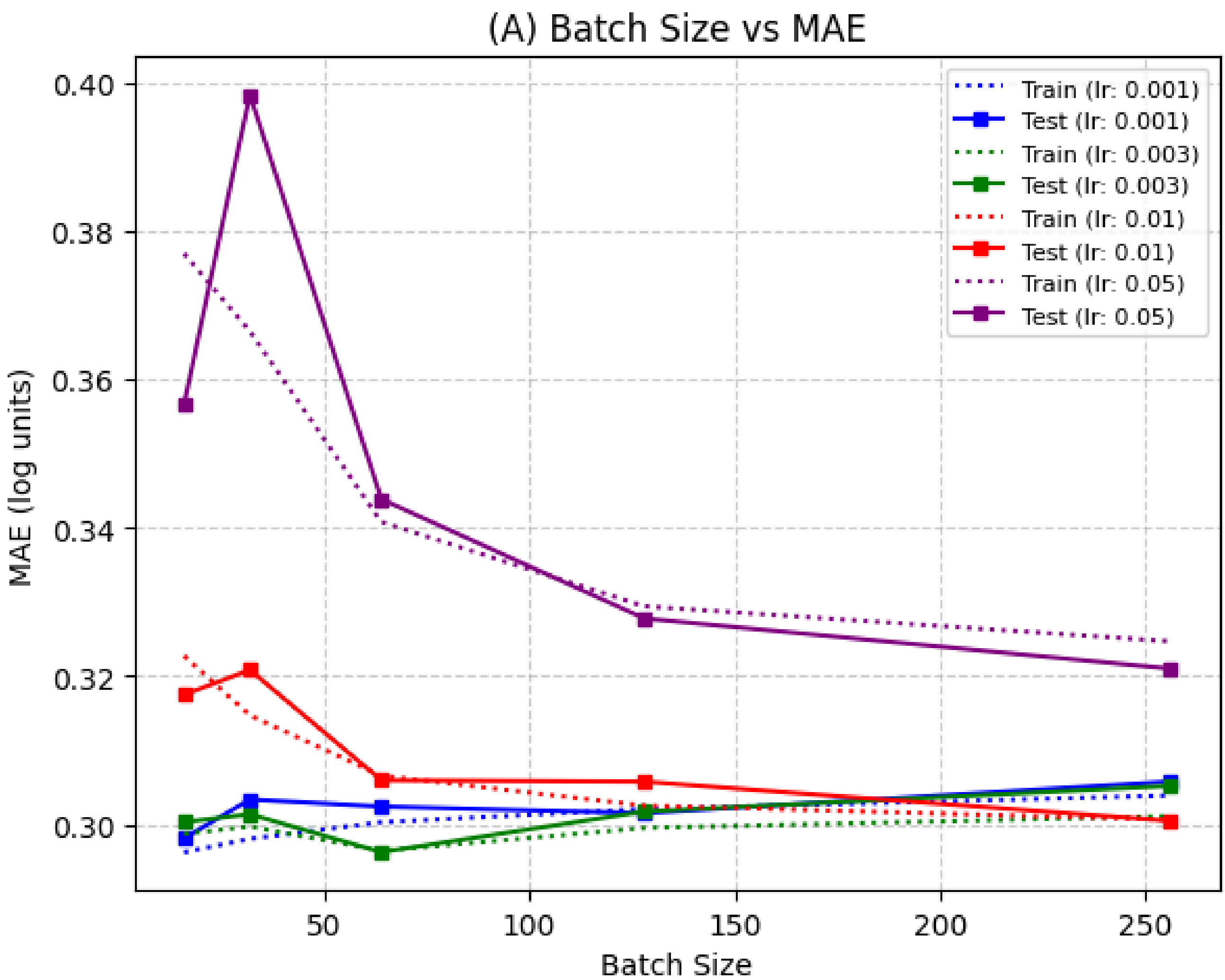
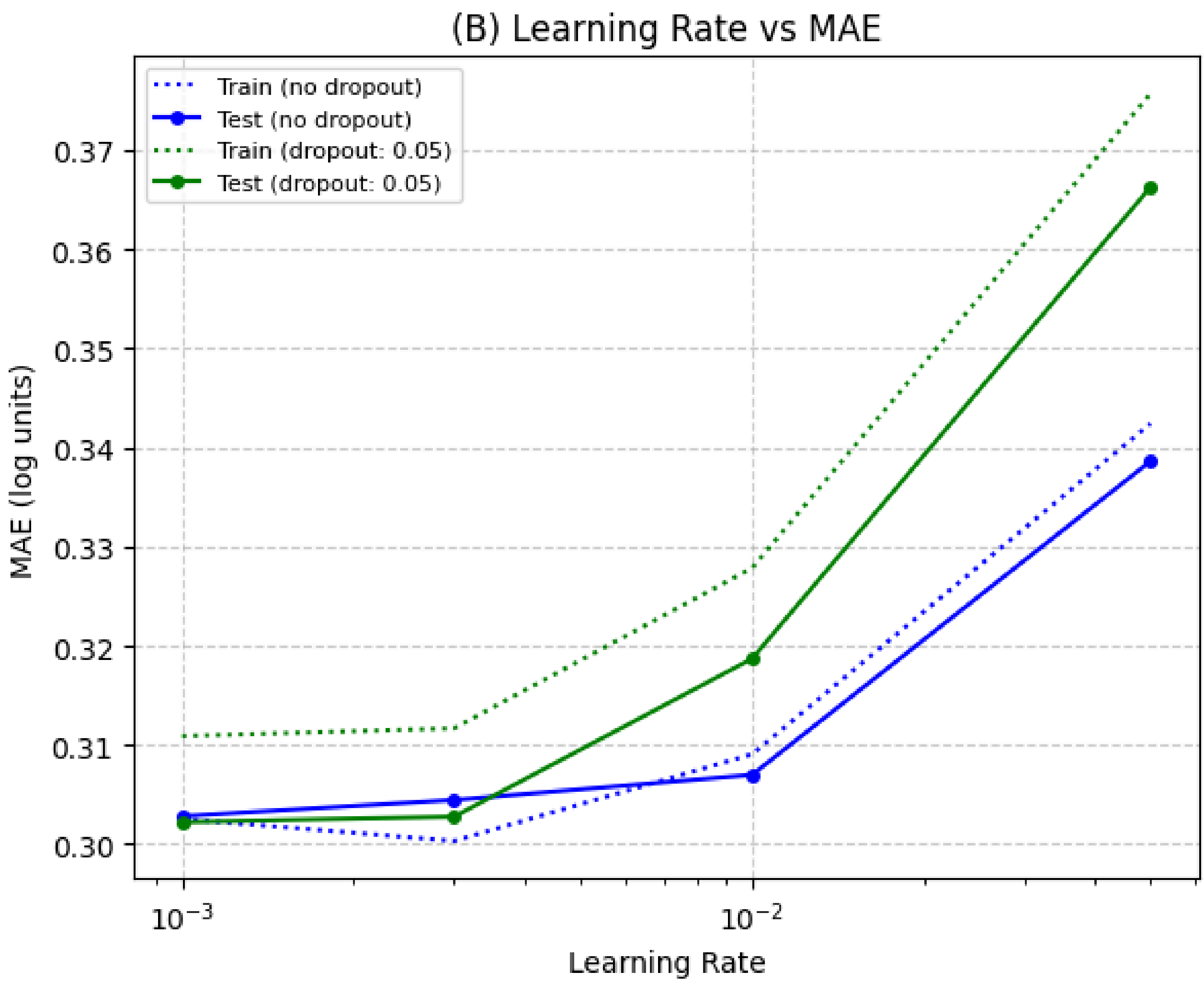
ANN Model

1. Hyperparameter Tuning



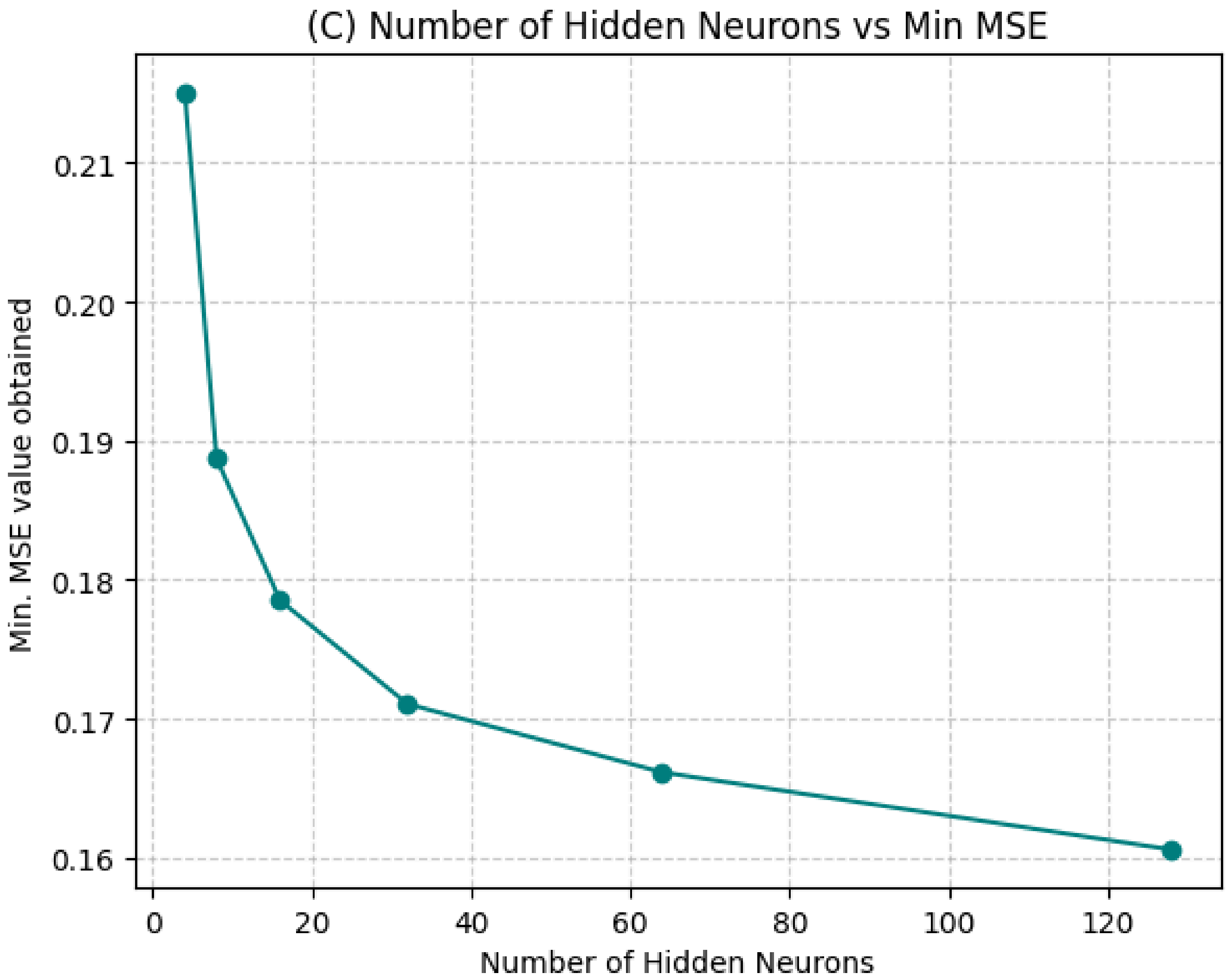
Plot (A): Batch Size vs. Mean Absolute Error (MAE)

- **Stability at Lower Learning Rates:** The model has stable convergence and lower errors when we use smaller learning rates (0.001 and 0.003).
- **Instability with High Learning Rates:** At a high learning rate of 0.05 , the model struggles to converge, particularly at smaller batch sizes. The error jumps significantly, indicating that the weight updates are too aggressive.
- **Optimal Batch Size:** For the best-performing learning rates, a batch size of 32 to 64 yields the lowest test MAE. It provides a good balance between accurate gradient estimation and training speed.



Plot (B): Learning Rate vs. Mean Absolute Error (MAE)

- **Preference for Smaller Learning Rates:** The model strongly prefers learning rates in the 0.001 to 0.003 range. As the learning rate increases toward 0.05, performance degrades.
- **Impact of Dropout:** Adding a dropout rate of 5% generally results in slightly worse performance compared to no dropout, especially at higher learning rates. Because this is a regression task built on physical parameters rather than a massive image dataset, the model is likely not suffering from severe overfitting, making dropout unnecessary or slightly detrimental.

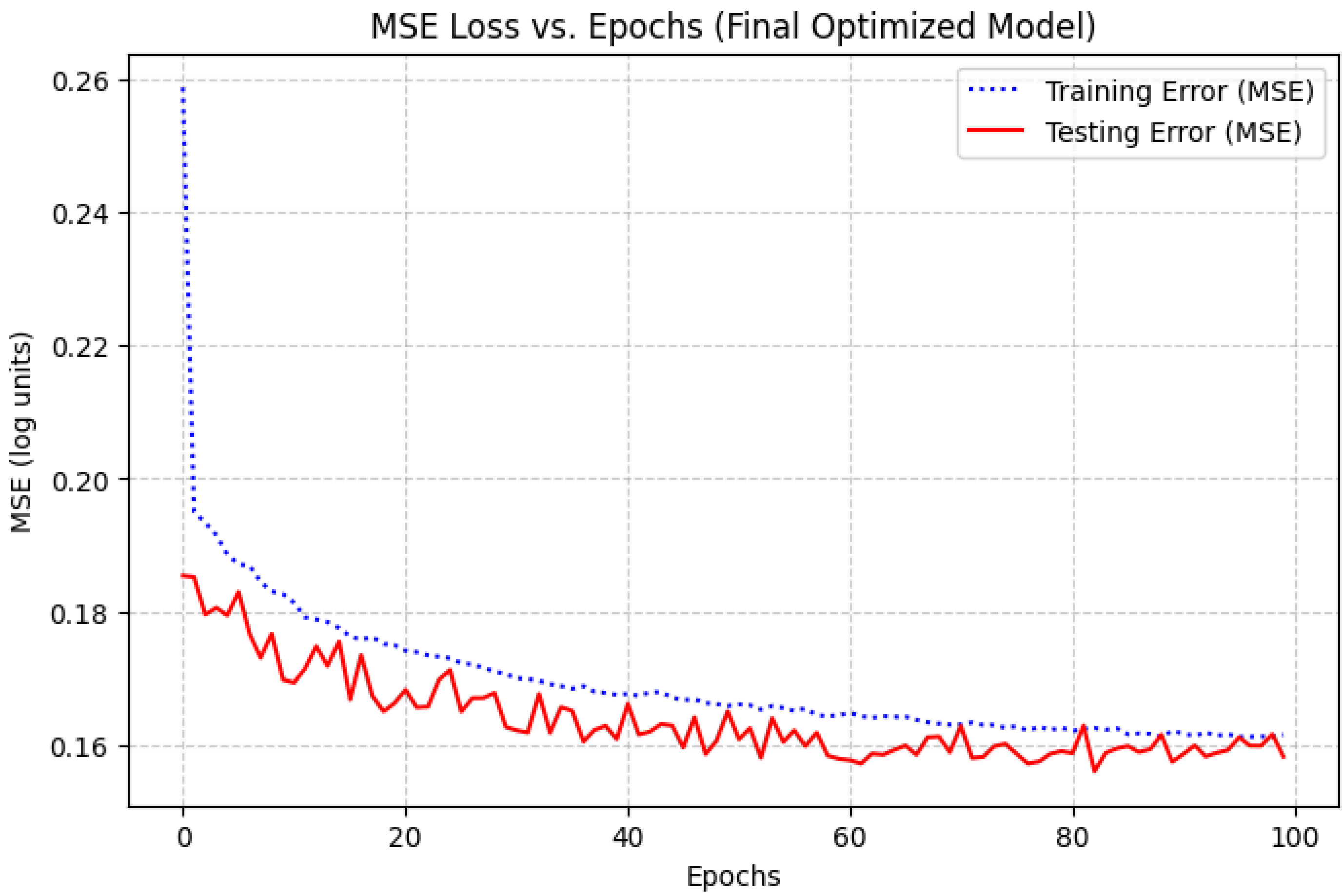


Plot (C): Number of Hidden Neurons vs. Minimum MSE

- There is a clear, inverse relationship between network capacity and error. As the number of hidden neurons increases from 4 to 32, the Minimum Mean Squared Error (MSE) drops sharply.
- Beyond 64 neurons, the curve begins to flatten. While 128 neurons achieve the absolute lowest MSE, the performance gain is marginal compared to the added computational complexity. Therefore, 64 or 128 neurons represent the optimal number for capturing the non-linear relationships in the seismic data.

Final Optimized Model Parameters

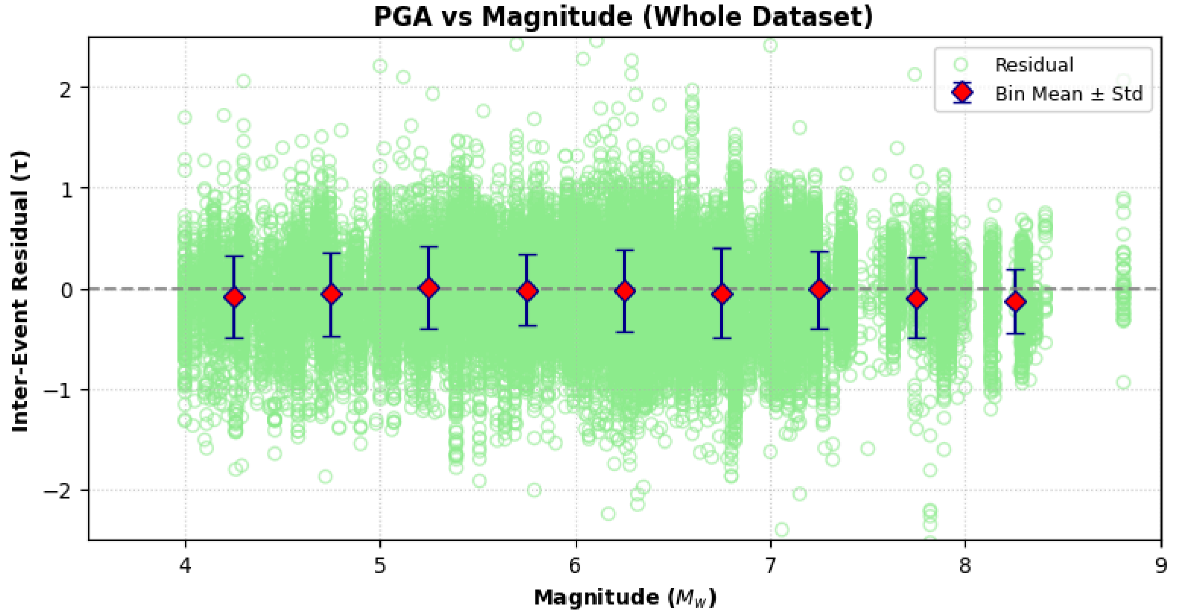
- Learning Rate: 0.001
- Batch Size: 32
- Number of Hidden Neurons: 128
- Dropout Rate: 0.0 (None)

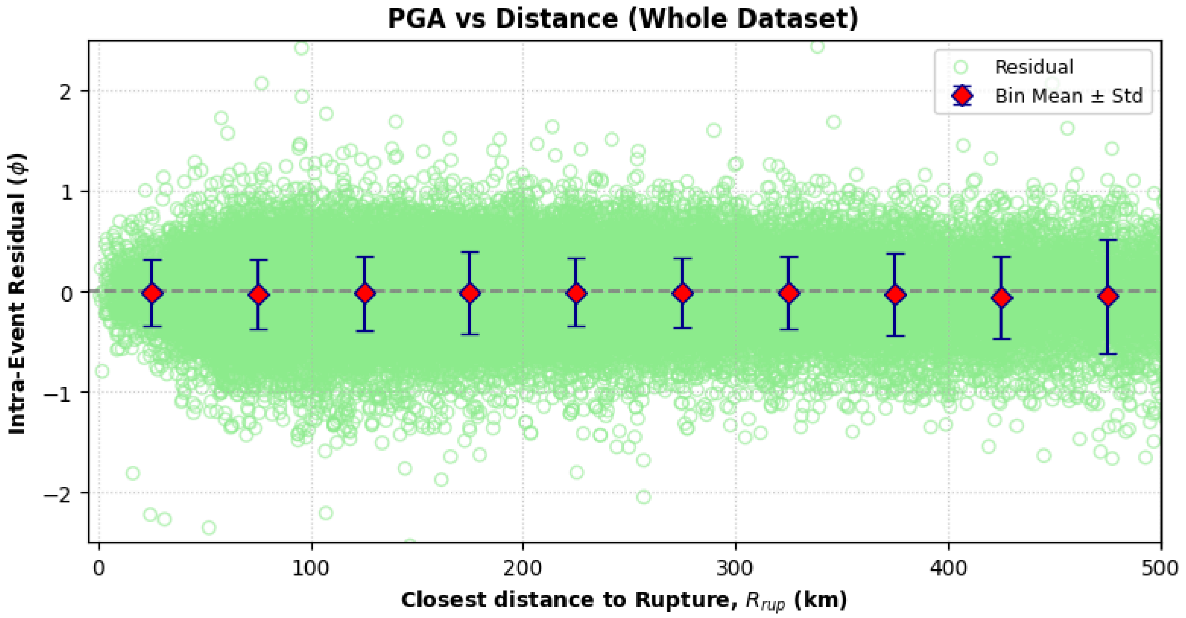


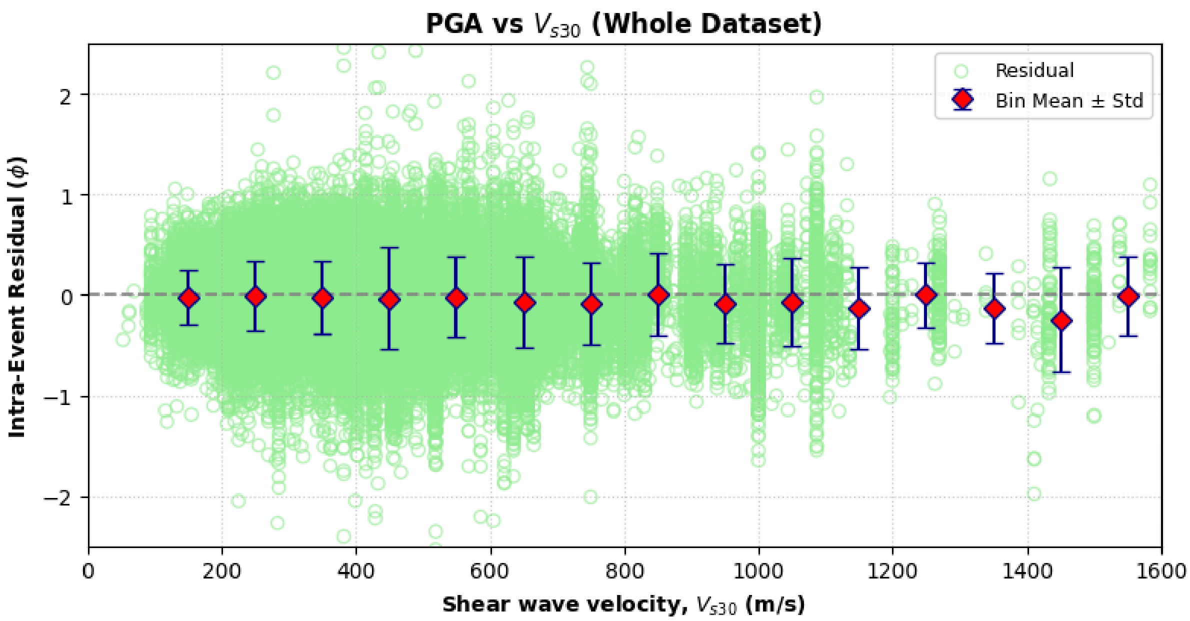
MSE Loss vs. Epochs

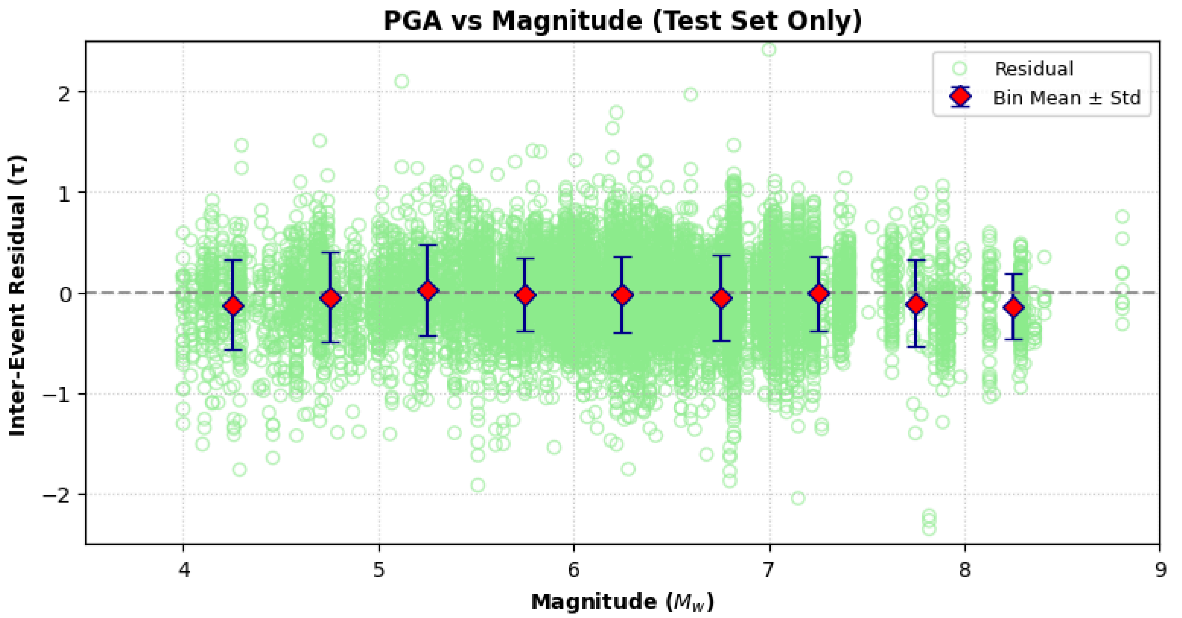
- The **Training Error (MSE)** drops sharply within the first 10 epochs, falling from approximately **0.26** to under **0.19**. This indicates that the chosen learning rate (0.001) effectively captures the primary physical trends early in the training.
- The **Testing Error (MSE)** closely tracks the training error throughout the 100 epochs. There is no widening gap between the blue dotted line and the red solid line, proving that the model generalizes well to unseen data without overfitting.
- Reaching a final MSE of ~0.16 confirms that the **128-neuron** was sufficient to minimize the global error.

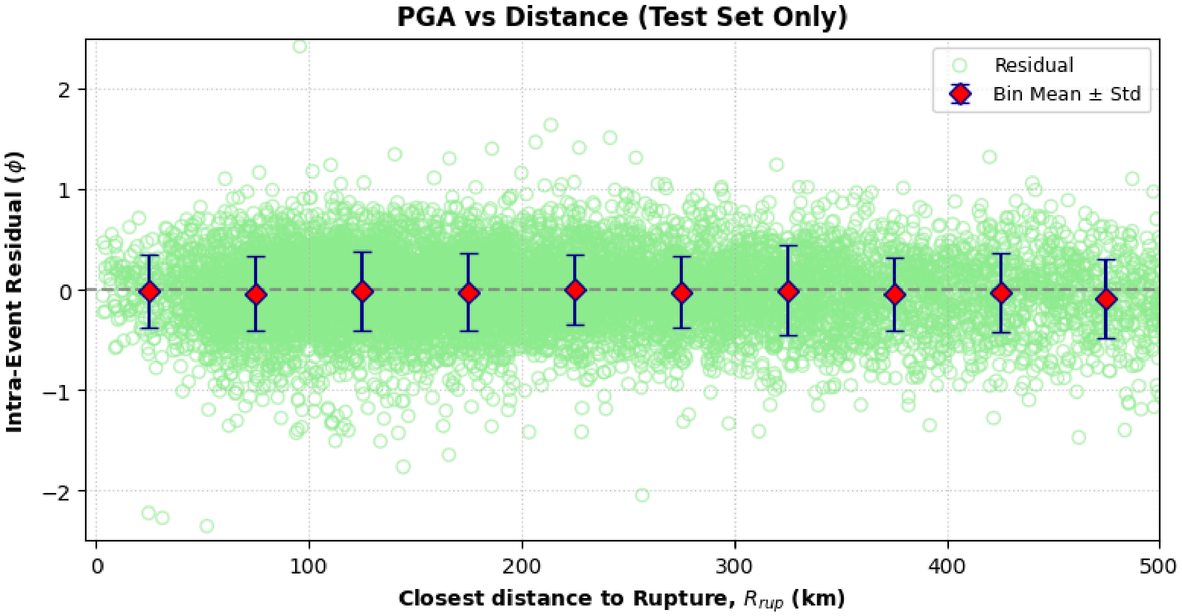
2. Residual Analysis

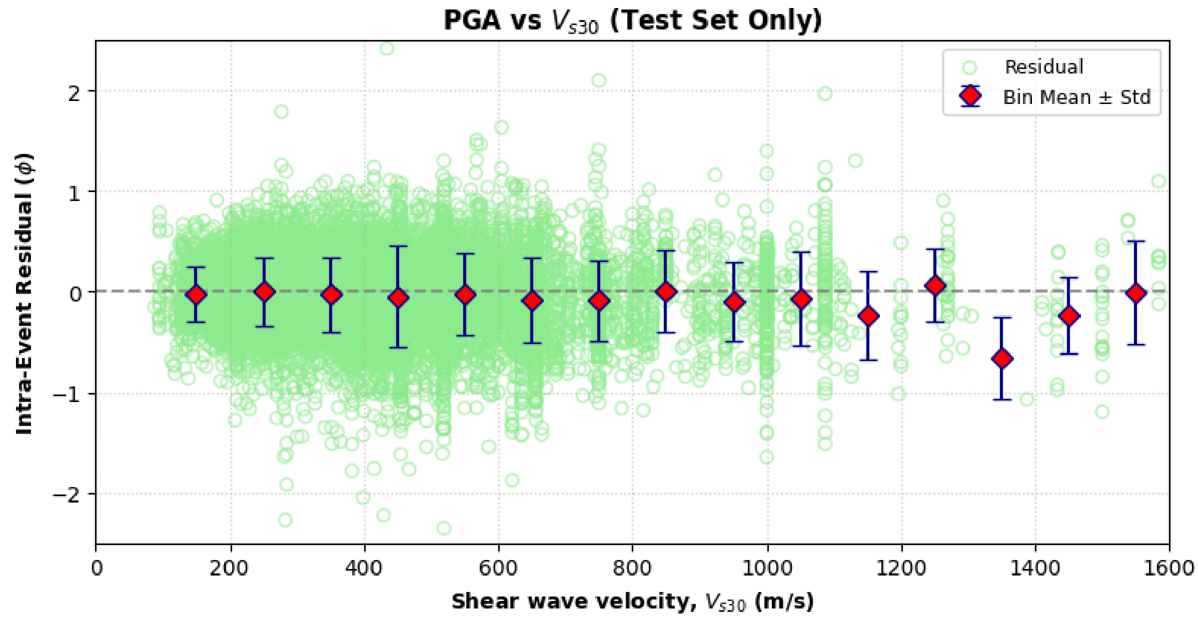







PGA vs Magnitude (Test Set Only)
Residual
Bin Mean ± Std
Inter-Event Residual (τ)
Magnitude (M_w)
2
1
0
−1
−2
4
5
6
7
8
9


PGA vs Distance (Test Set Only)
Residual
Bin Mean ± Std
Intra-Event Residual (ϕ)
Closest distance to Rupture, R_{rup} (km)
2
1
0
−1
−2
0
100
200
300
400
500


PGA vs V_{s30} (Test Set Only)
Residual
Bin Mean ± Std
Intra-Event Residual (ϕ)
Shear wave velocity, V_{s30} (m/s)
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400
1600

1. Residuals vs. Magnitude (M_w)

- For both the whole dataset and the test set, the binned mean residuals (red diamonds) consistently hover near the zero-line. This indicates that the model is unbiased and correctly predicts shaking intensity for both small (M_w~4) and large (M_w~8) earthquakes without systematic under- or over-estimation.
- The consistent spread of residuals across different magnitudes confirms that the model's performance is stable regardless of the earthquake's size.

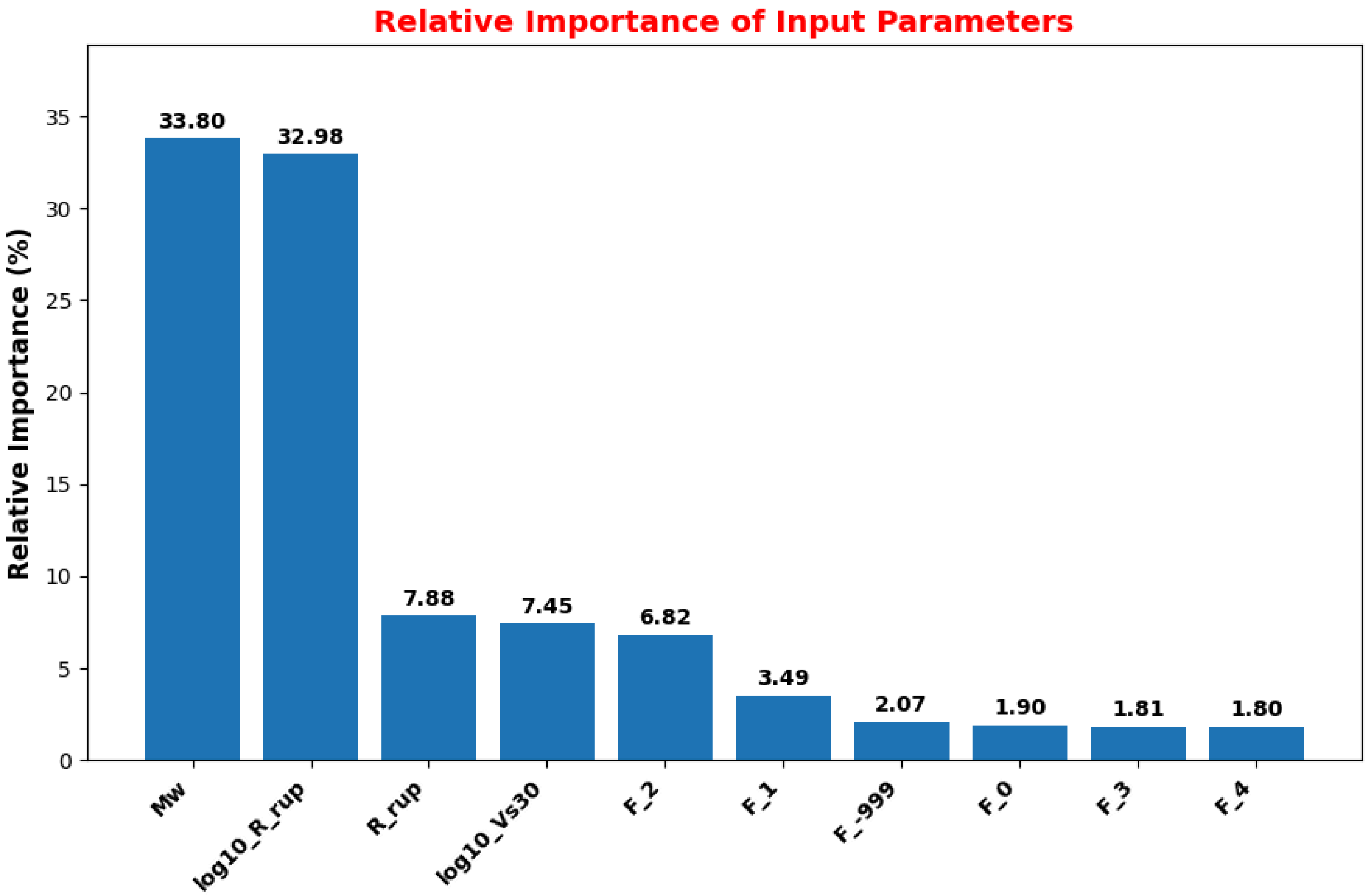
2. Residuals vs. Distance (R_{rup})

- The binned means stay centered at zero throughout the distance range from 0 to 500 km. This proves that our model has successfully learned the physical law of seismic attenuation, how ground motion intensity decreases as distance from the rupture increases.
- At closer distances (0–200 km), where data is most dense, the error bars are very tight, showing high model confidence in high-risk zones.

3. Residuals vs. Shear Wave Velocity (V_{s30})

- The model shows excellent performance across different soil conditions, from soft sites (V_{s30} < 200 m/s) to hard rock (V_{s30}> 1000 m/s). The binned means remain near zero, suggesting the model accurately accounts for site amplification effects.
- At very high velocities (V_{s30} > 1200 m/s), we can observe slightly wider error bars and a minor dip in the test set mean. This is expected due to the relative scarcity of hard-rock recordings in subduction datasets compared to soil sites.

3. Feature Importance

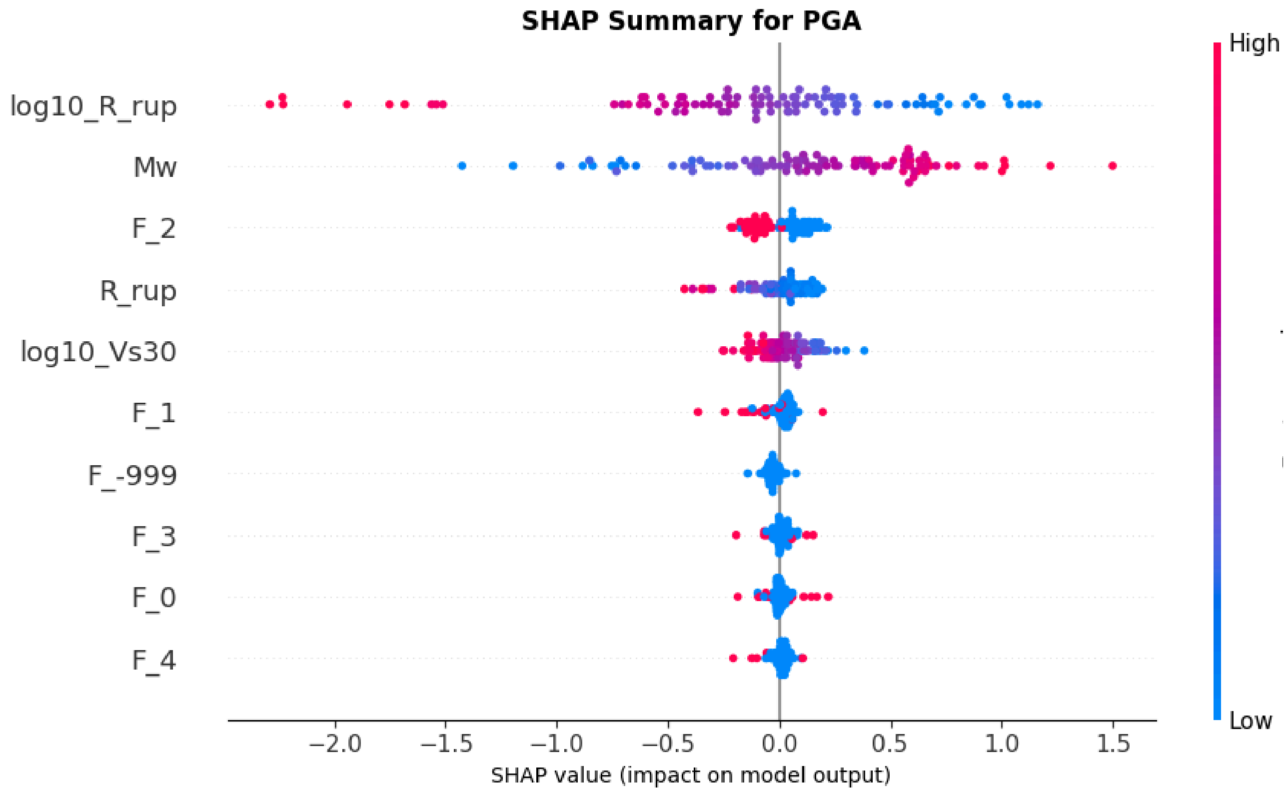


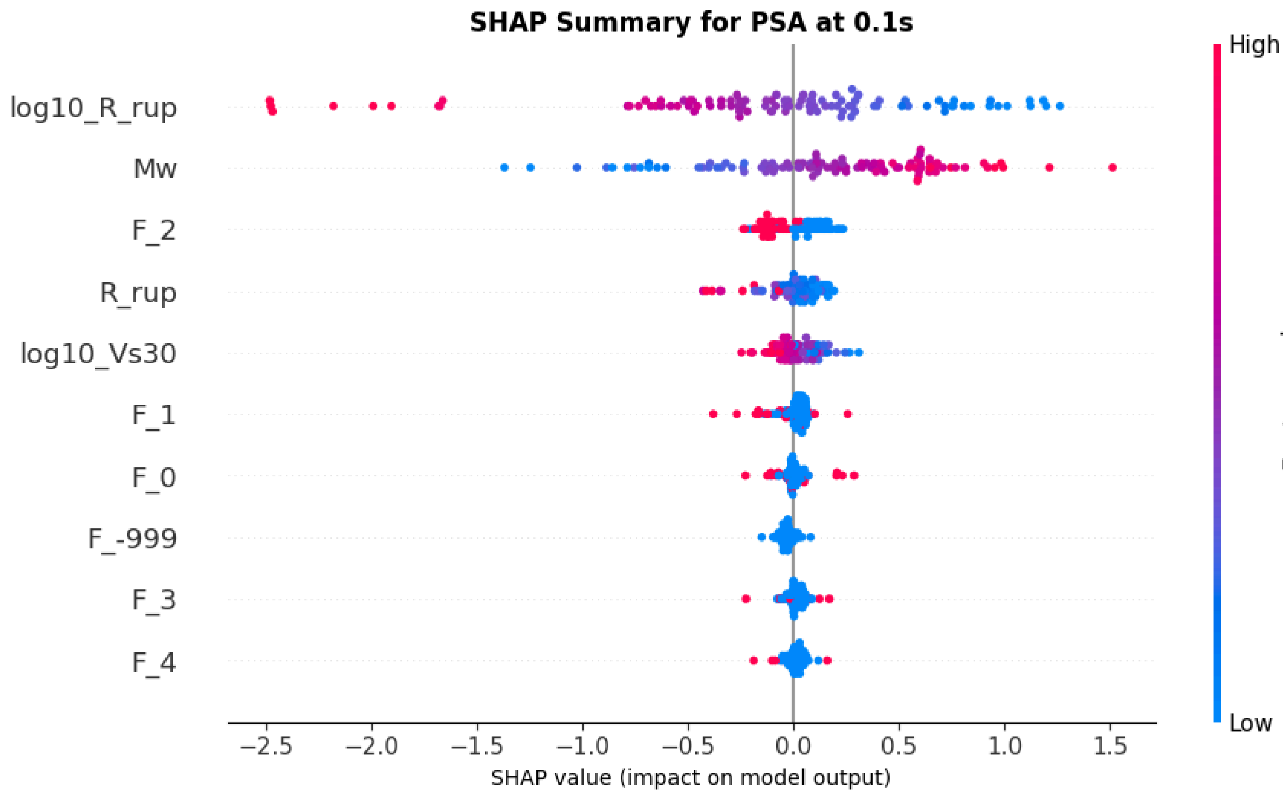
Bar Plot: Global Feature Importance

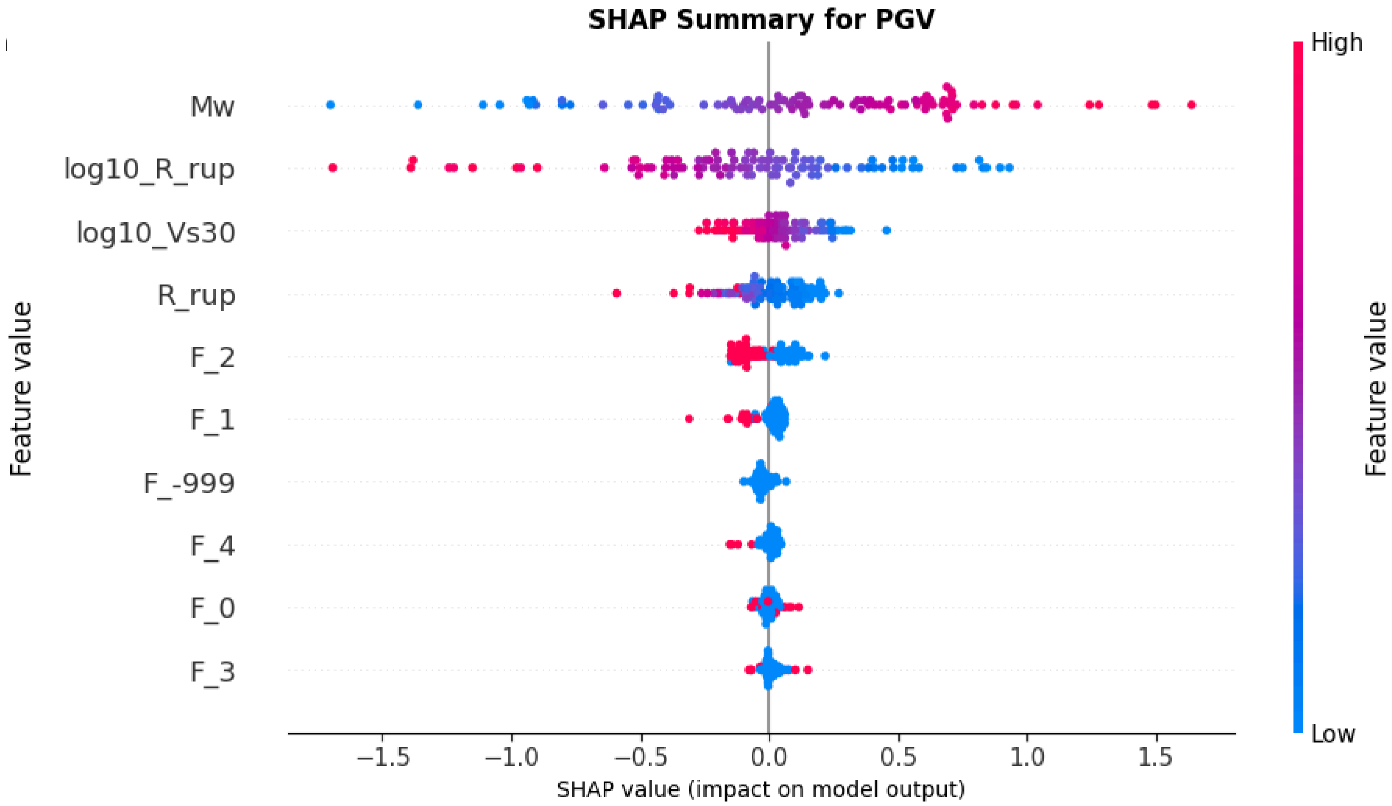
The bar chart quantifies the contribution of each input parameter to the model's predictive power:

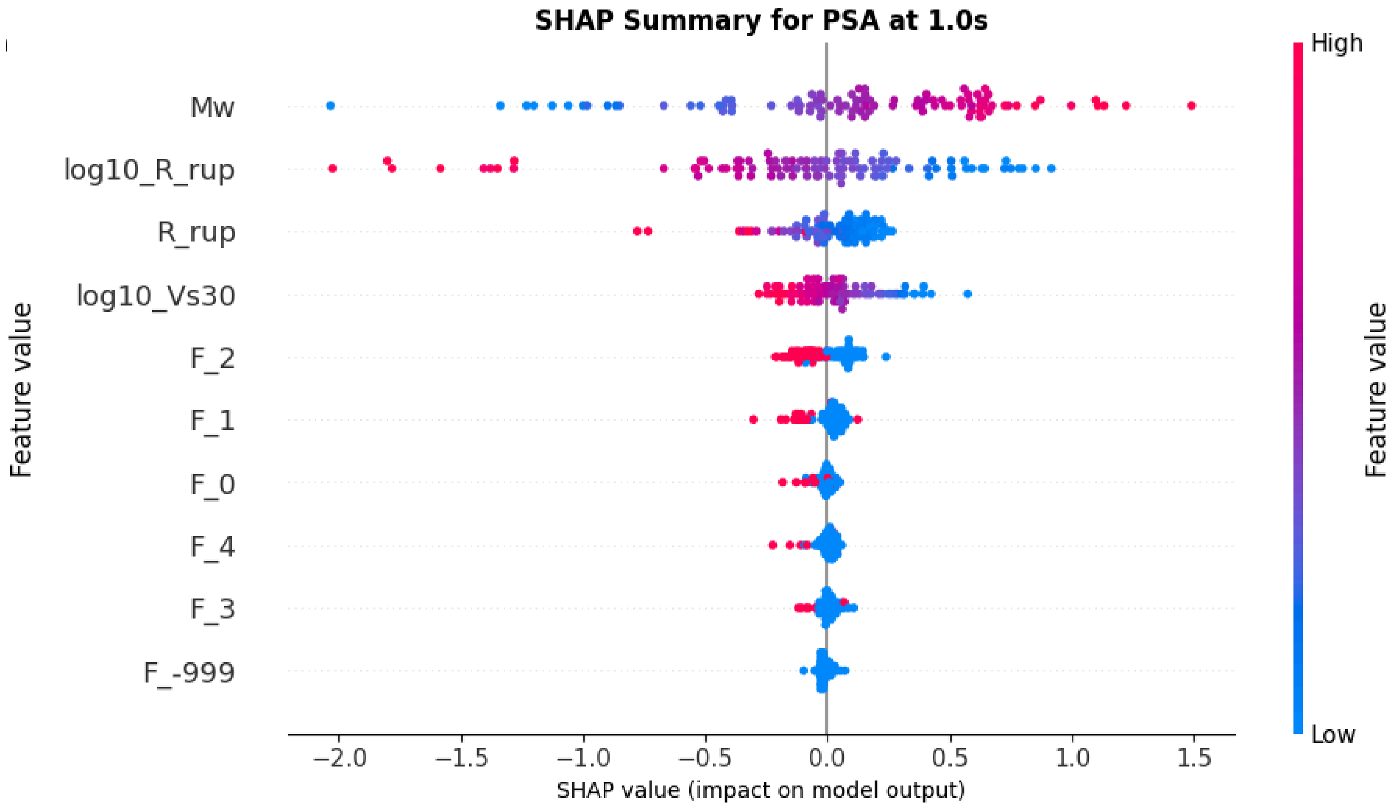
- Earthquake Magnitude (**Mw**) and the log-transformed distance (**$\log_{10}R_{rup}$**) are the dominant features, contributing approximately **33.80%** and **32.98%** respectively. This aligns perfectly with seismic theory, where magnitude and distance are the fundamental controllers of ground motion intensity.
- The soil condition, represented by **$\log_{10}Vs30$**, holds a significant importance of **7.45%**. This demonstrates that the model successfully prioritizes site amplification effects.
- Raw distance (**R_rup**) and specific Fault Types (**F_2, F_1, etc.**) have lower relative importance, indicating they provide auxiliary refinements rather than primary trends.

4. SHAP Analysis






SHAP Summary for PGV
Mw
log10_R_rup
log10_Vs30
R_rup
F_2
F_1
F_-999
F_4
F_0
F_3
Feature value
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
SHAP value (impact on model output)
High
Low
Feature value

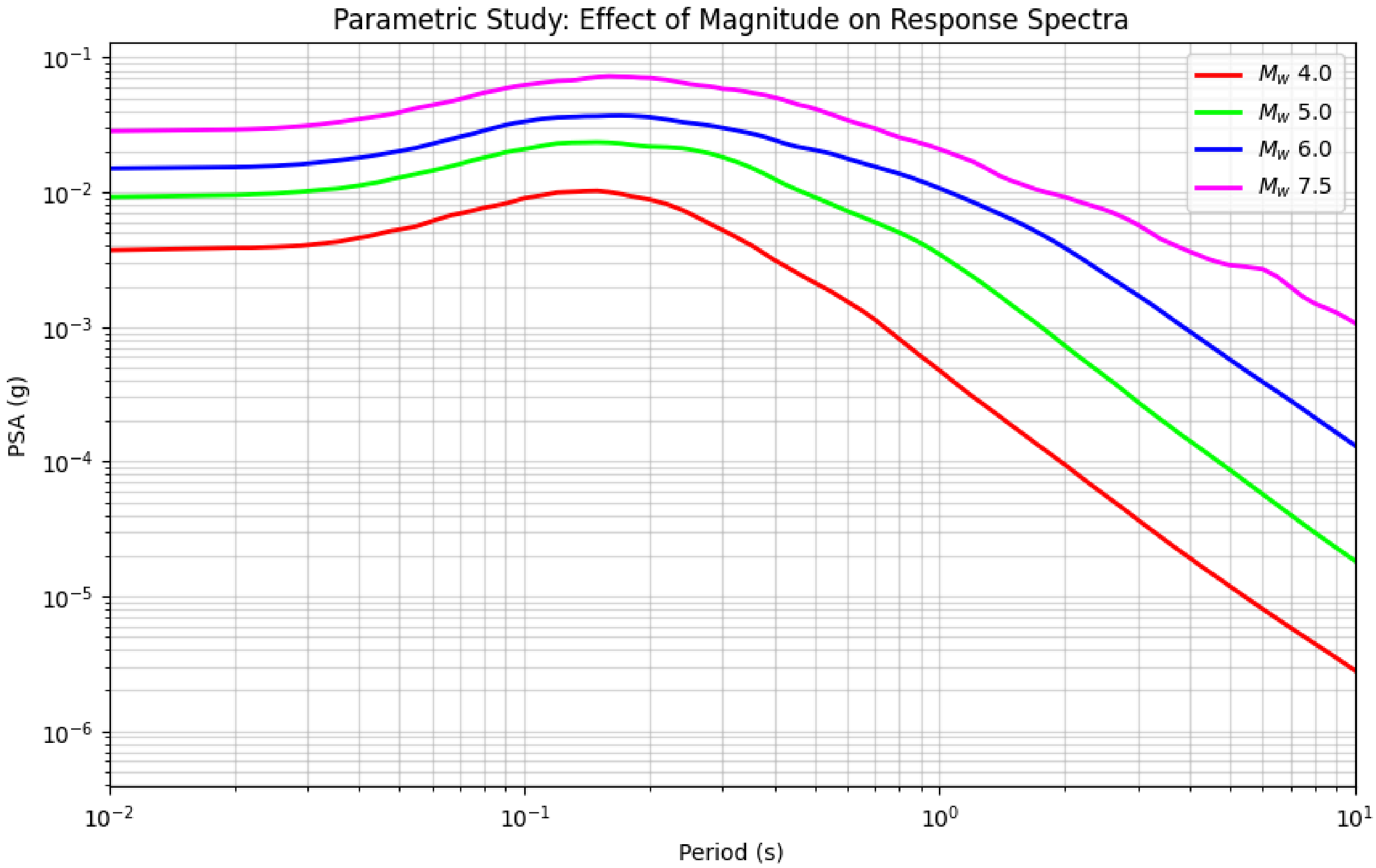

SHAP Summary for PSA at 1.0s
Mw
log10_R_rup
R_rup
log10_Vs30
F_2
F_1
F_0
F_4
F_3
F_-999
Feature value
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
SHAP value (impact on model output)
High
Low
Feature value

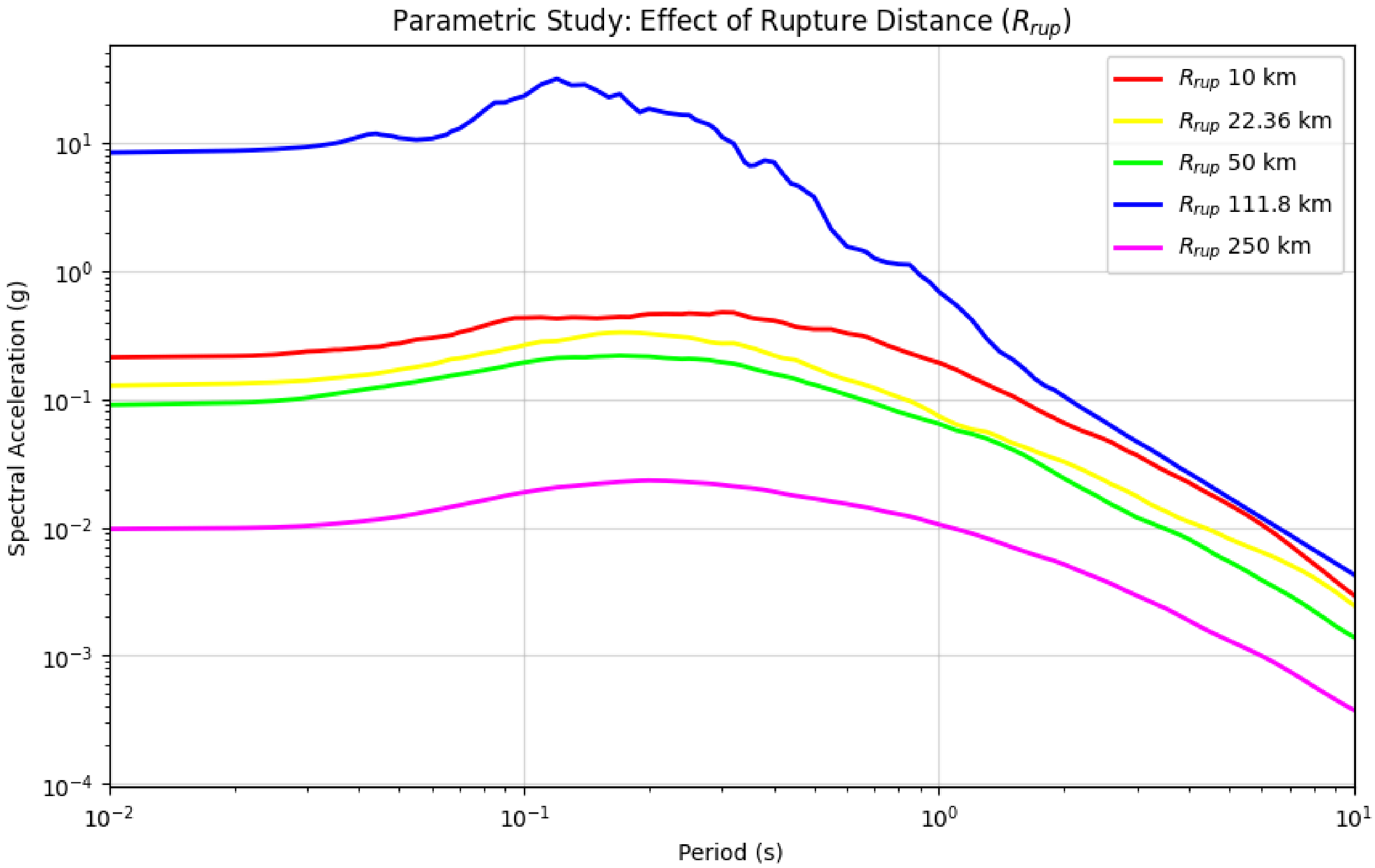
Beeswarm Plots: SHAP Analysis

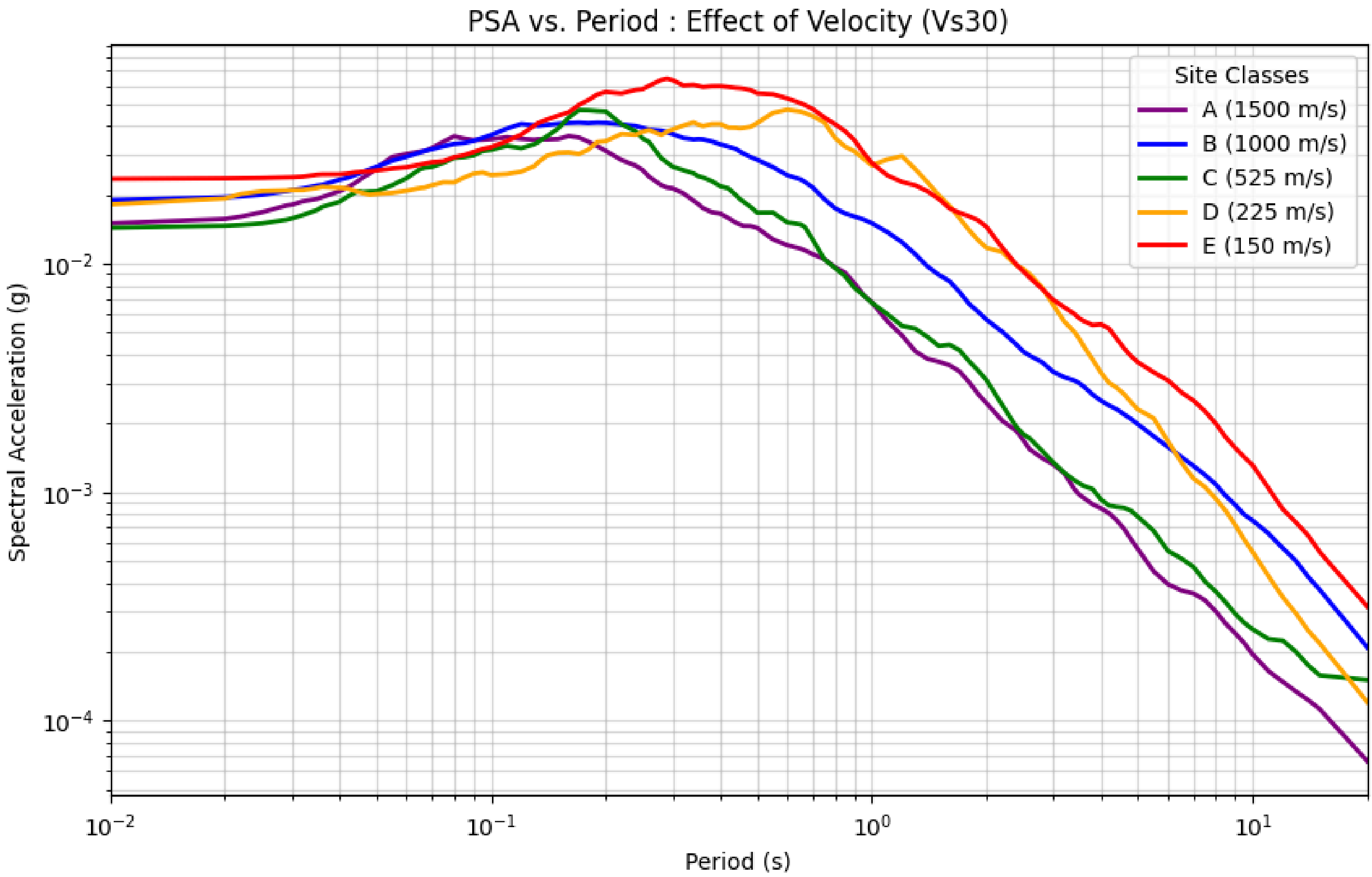
The SHAP beeswarm plots show the **direction** of the impact for specific Intensity Measures (PGA, PGV, etc.):

1. **Directional Impact**:
    a. **Magnitude (High Values in Red)**: Red dots consistently appear on the right side of the center line (>0.0). This confirms that higher magnitudes physically push the model to predict higher shaking intensity.
    b. **Distance (High Values in Red)**: For distance-related features, red dots appear on the left side (<0.0). This visually validates **attenuation**, where increasing distance reduces the impact on the model output.
2. **Frequency Sensitivity**:
    a. Comparison between the **PSA at 0.2s** and **PSA at 1s** summaries shows that different features have varying weights depending on the period. For instance, certain features (like Feature 7 or Feature 4) show a much wider spread for short-period (0.2s) vs. long-period (1s) structures.

5. Parametric Study







Parametric Study: Insights

1. Effect of Magnitude (M_w) on Response Spectra

- As earthquake magnitude increases from M_w 4.0 to 7.5, the Spectral Acceleration (PSA) shifts upward across all periods. This is a clear validation that the model correctly predicts higher energy release for larger events.

2. Effect of Rupture Distance (R_{rup})

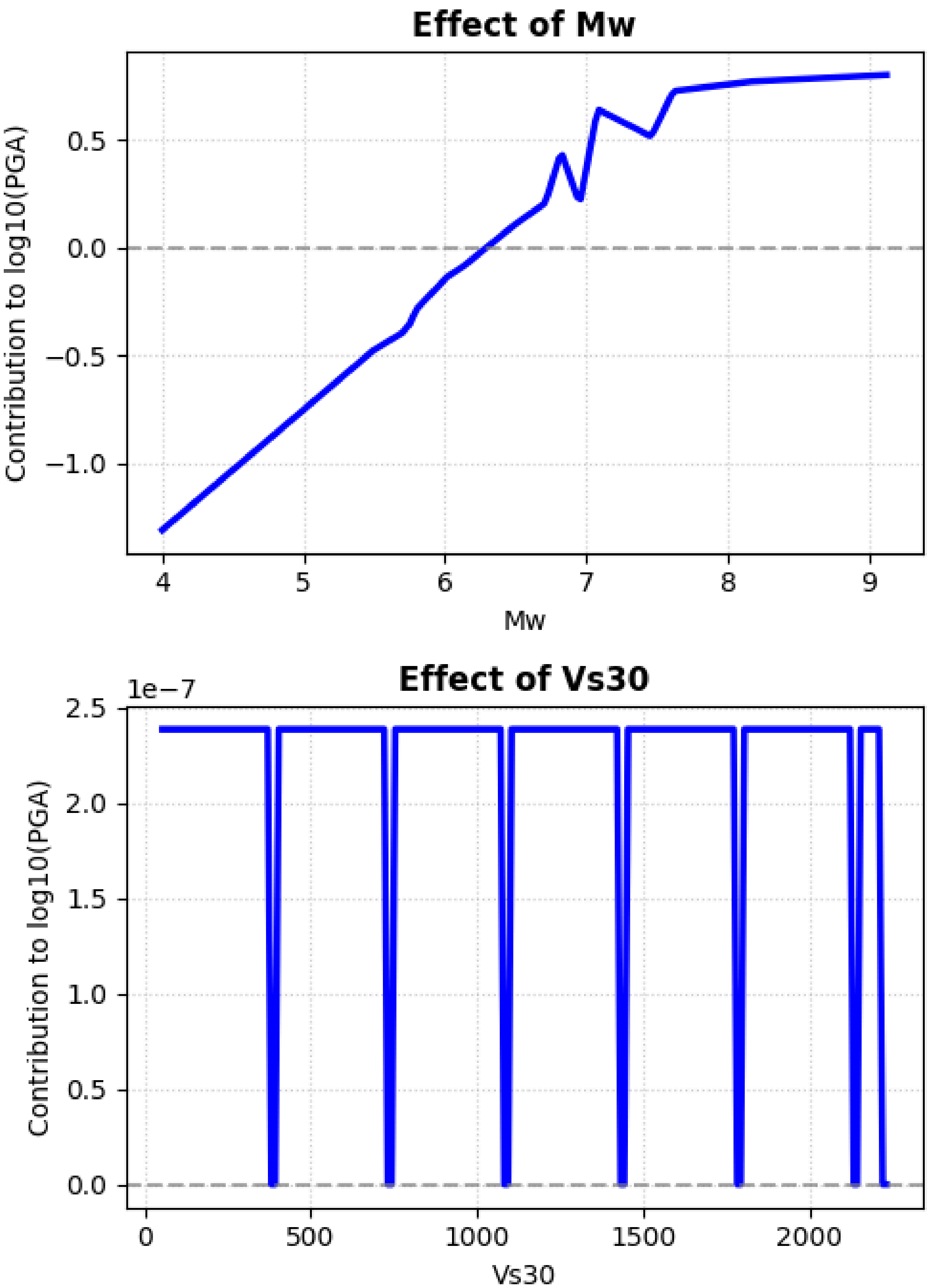
- Shaking intensity decreases significantly as the distance from the rupture increases from 10 km to 250 km. The magenta line ($R_{rup=250km}$) sits orders of magnitude lower than the red line ($R_{rup=10km}$)), confirming the model accurately simulates seismic wave attenuation.

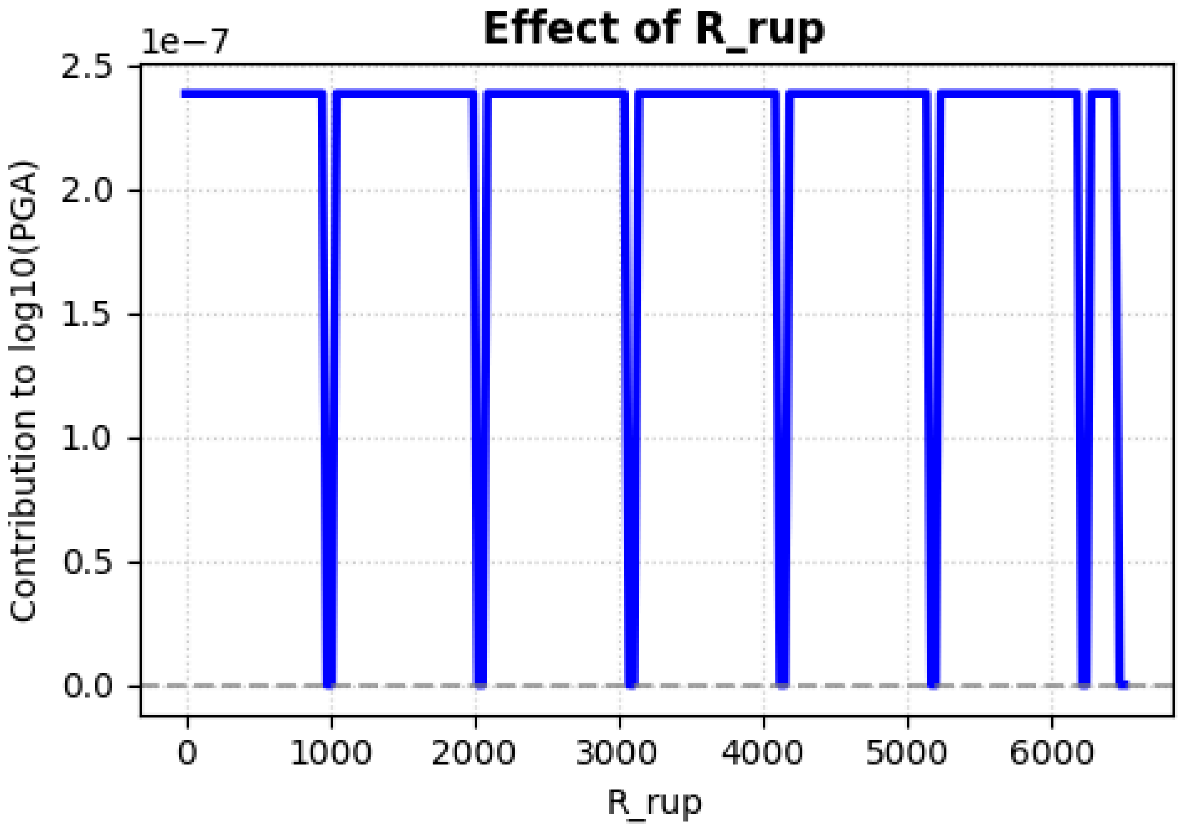
3. Effect of Site Velocity (V_{s30})

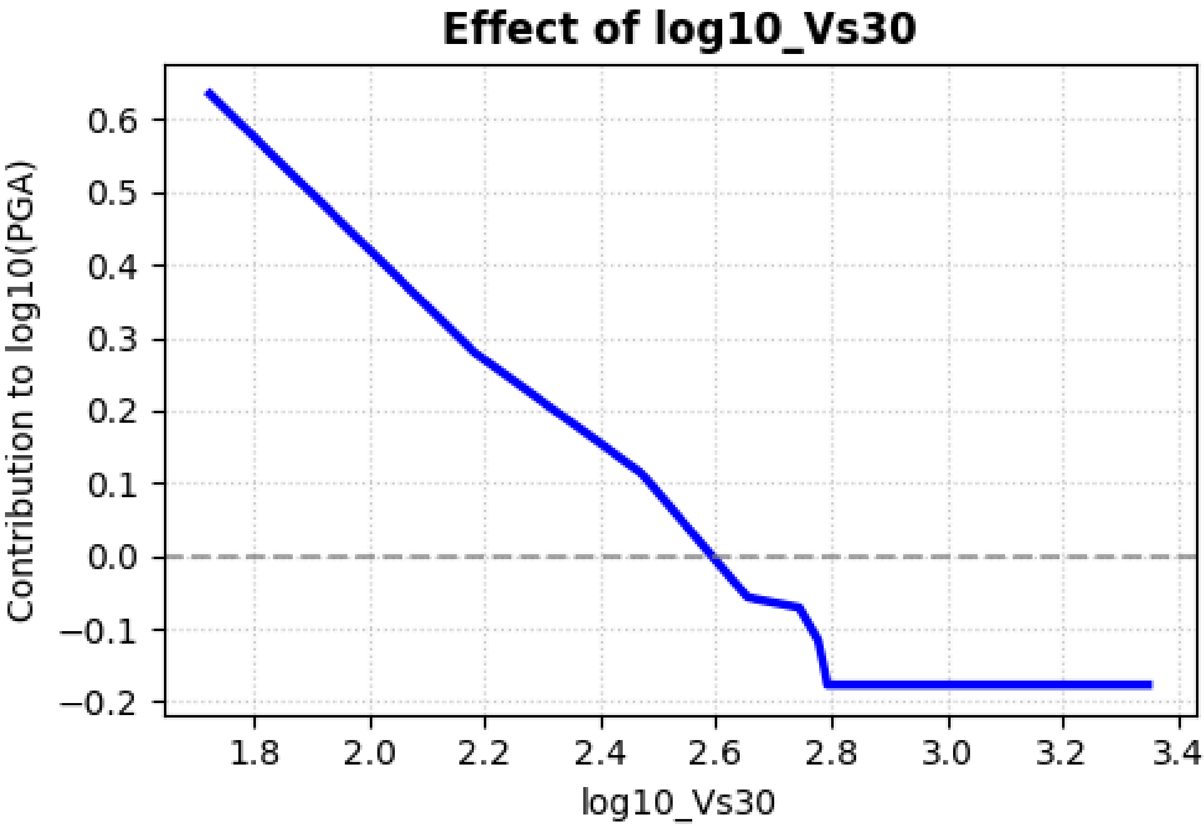
- The plot clearly distinguishes between different Site Classes. Class E (soft soil, 150 m/s in red) shows the highest spectral acceleration, particularly at longer periods, due to soil amplification.
- Conversely, Class A (hard rock, 1500 m/s) exhibits the lowest PSA levels. This confirms the model has correctly learned that denser, stiffer materials dissipate seismic energy more effectively than loose soils.

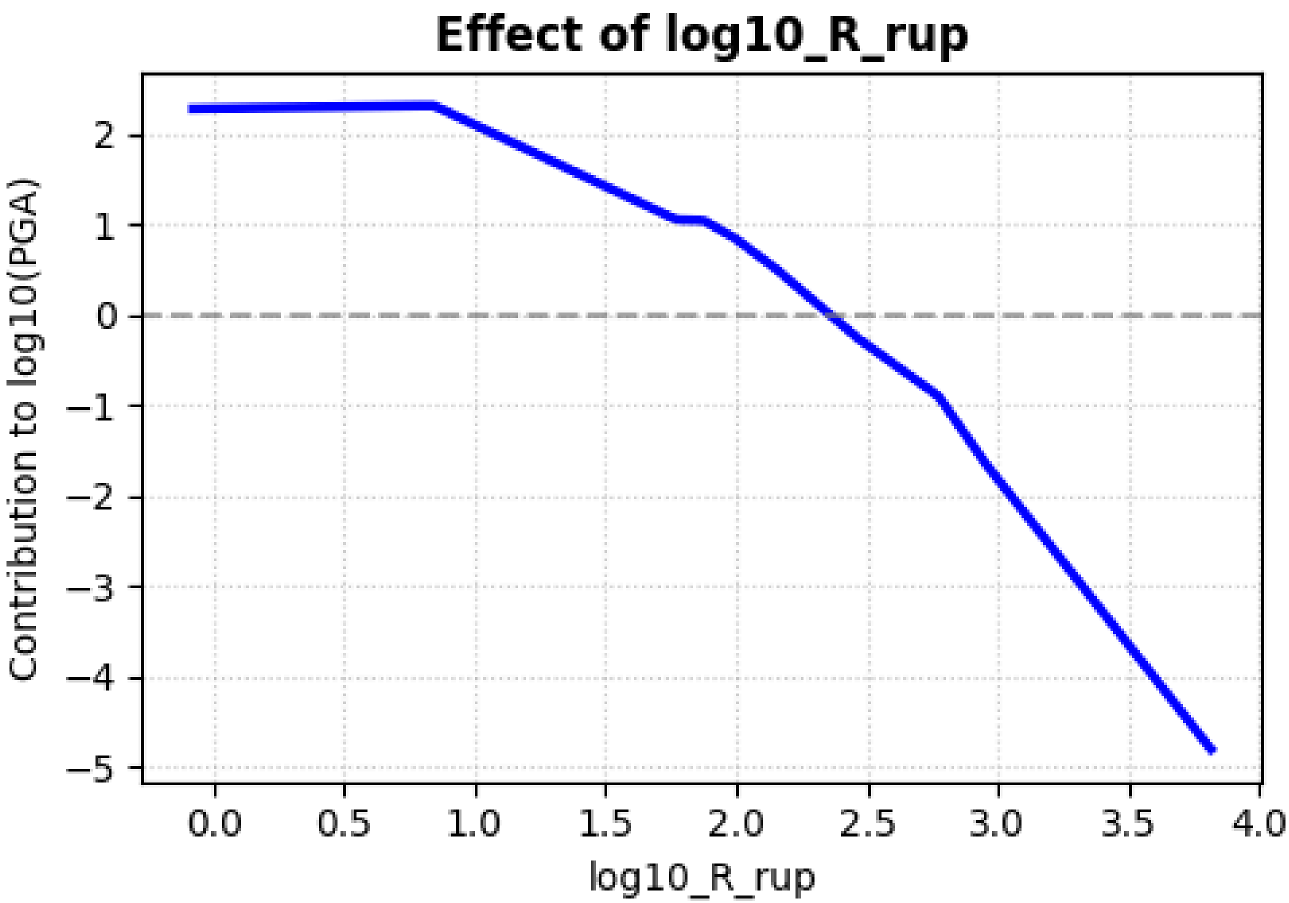
Neural Additive Model (NAM)

<u>NAM Learned Shape Functions for PGA</u>




Effect of R_rup
1e−7
2.5
2.0
1.5
1.0
0.5
0.0
Contribution to log10(PGA)
0
1000
2000
3000
4000
5000
6000
R_rup


Effect of log10_Vs30
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
Contribution to log10(PGA)
1.8
2.0
2.2
2.4
2.6
2.8
3.0
3.2
3.4
log10_Vs30



NAM Shape Functions: Insights

1. Physical Trend Validation

- **Effect of Mw (Magnitude)**: The model learns a clear **positive correlation**, where the contribution to $\log_{10}$PGA increases steadily from Magnitude 4 to approximately 7.5. Beyond M_w 7.5, the curve begins to flatten, successfully capturing the physical phenomenon of **magnitude saturation**.
- **Effect of $\log_{10}R_{rup}$ (Distance)**: The shape function shows a sharp, linear **downward trend**. This confirms that as the logarithmic distance increases, the ground motion intensity **attenuates** significantly, adhering to the inverse-square law of seismic energy dissipation.
- **Effect of $\log_{10}V_{s30}$ (Soil Stiffness)**: The model captures **site amplification**. The contribution to PGA is high for lower velocities (soft soil) and drops as velocity increases toward stiffer rock sites (higher $\log_{10}V_{s30}$ values).

2. Automatic Feature Selection

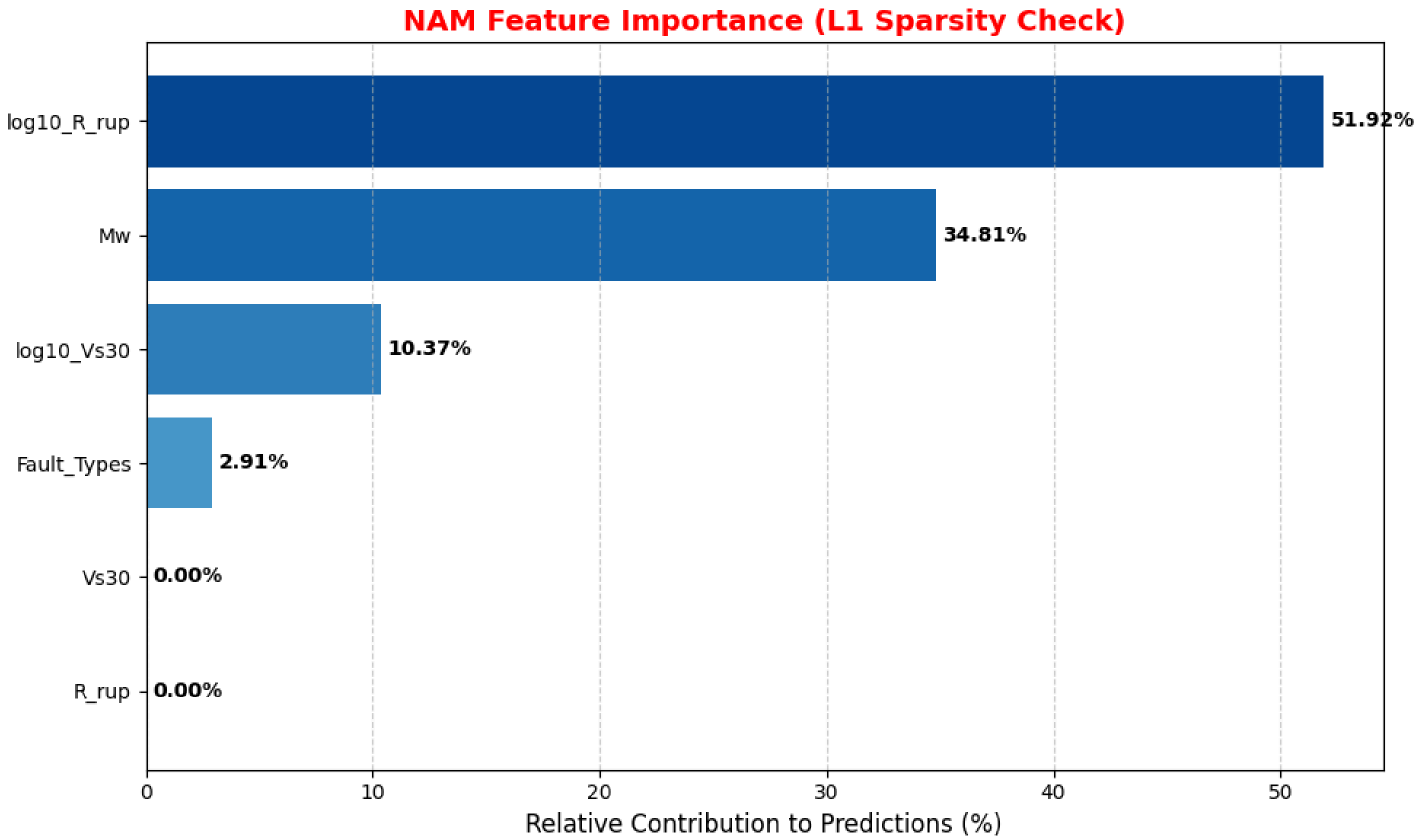
I. **Raw vs. Log Parameters**: One of the most powerful insights from NAM is the **L1 Sparsity check**.

   A. The shape functions for raw R_{rup} and raw V_{s30} are essentially flat lines at zero

   B. This indicates that the model's L1 regularization **automatically "turned off"** the raw versions of these parameters in favor of their log-transformed counterparts.

   C. The model mathematically determined that $\log_{10}R_{rup}$ and $\log_{10}V_{s30}$ are the superior representations for predicting ground motion.

NAM Feature Importance



NAM Feature Importance : Insights:

- **Dominance of Log-Distance**: Log-transformed distance (**log10_R_rup**) is the most critical feature, accounting for **51.92%** of the model's predictive weight. This confirms that seismic attenuation is best captured on a logarithmic scale.
- **Magnitude Impact**: Earthquake Magnitude (**Mw**) is the second most vital parameter at **34.81%**, reflecting its fundamental role in determining ground motion energy.
- **Site Amplification**: Soil stiffness (**log10_Vs30**) contributes **10.37%**, showing that while Magnitude and Distance are primary, site-specific soil conditions remain a significant secondary factor.
- **Automated Feature Selection**: The L1 regularization successfully "zeroed out" the raw **Vs30** and **R_rup** variables (**0.00% importance**), proving the model automatically selected log-transforms as the mathematically superior inputs.
- **Fault Mechanism Influence**: Categorical **Fault_Types** have a minor influence (**2.91%**), suggesting that rupture style provides a small refinement to the final intensity prediction.